

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

 show ip route ospf

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network?

What is the accumulated cost metric for the 192.168.2.0/24 network on R3?

Does R2 show up as an OSPF neighbor on R1?

Does R2 show up as an OSPF neighbor on R3?

What does this information tell you?

 It shows that R2 is an OSPF neighbor of both R1 and R3 and acts as the intermediate router between them, so routes to 192.168.2.0/24 traverse R2 and their total cost includes the serial links on both sides of R2.

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

 R2(config)# router ospf 1

R2(config-router)# no passive-interface s0/0/1

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated? 64 , because R3 now reaches 192.168.2.0/24 directly via a single serial link to R2, so only one 64-cost serial interface is counted.

Is R2 listed as an OSPF neighbor to R3? Yes

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

Because the default of 100 Mbps makes all interfaces at 100 Mbps or higher have the same cost of 1, so you cannot distinguish FastEthernet, Gigabit, and 10G links; increasing the reference bandwidth lets OSPF assign different costs to high-speed links and choose paths that more accurately reflect real bandwidth.

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

For 192.168.3.0/24: R1 uses S0/0/1 (cost 64) to reach R3, then R3's G0/0 (cost 1) to reach the LAN, so the total cost is $64 + 1 = 65$.

For 192.168.23.0/30: R1 has two equal-cost paths via S0/0/0 and S0/0/1, and each serial hop on those paths has cost 64; the accumulated cost for each route is therefore $64 + 64 = 128$, which is why two equal-cost routes appear.

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

The new accumulated cost is about 156, because with reference-bandwidth 10000, a 128 Kbps serial link has cost $10000 / 128 \approx 78$, and the path to 192.168.23.0/30 still traverses two such serial links, so their costs add up to roughly $78 + 78$.

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

After changing bandwidth or explicit OSPF costs, the total cost of the path from R1 to 192.168.3.0/24 via R2 becomes lower than the cost of the direct path via R3, so OSPF selects the lower-cost route and forwards traffic through R2.

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

Because the router ID must be unique in the OSPF domain and is used in neighbor relationships, LSA origination, and DR/BDR elections; manually controlling it prevents changes when interface IPs change and makes troubleshooting and topology interpretation more stable and predictable.

2. Why is the DR/BDR election process not a concern in this lab?

Because the main topology uses point-to-point serial links where DR/BDR are not used and only a small LAN segment with few routers, so DR/BDR elections do not significantly affect convergence or the main learning objectives of the lab.

3. Why would you want to set an OSPF interface to passive?

To advertise the network without sending or receiving OSPF Hello packets on that interface, which reduces unnecessary routing traffic on host-only LANs and improves security by preventing unintended neighbor adjacencies.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

OSPF's use of **areas** (with backbone area 0 and multiple non-backbone areas) makes it hierarchical, allowing route summarization at area borders and limiting LSDB size and flooding scope so large topologies remain scalable.

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

network 10.0.0.0 0.255.255.255 area 0

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

The EIGRP route is used, because its administrative distance (90) is lower than OSPF (110) and RIP (120), and administrative distance is considered before the metric when choosing between different routing protocols.